

DRY-RUN GUIDE | suffix_sum.py
Source: Python\4th_belt\suffix_sum.py

CORE BEHAVIOR

Builds cumulative sums from right to left.

REPRESENTATIVE INPUT

[2,1,3,4].

STEP TRACE

From right: 4, 7, 8, 10; place them at original indices.

EXPECTED OUTPUT / RETURN

[10,8,7,4]

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\4th_belt\suffix_sum.py; line numbers refer to the original file. No source was modified.

001 | # Question: Compute the suffix sum of each element in the array.

002 | # Add your solution here.

003 |

004 | # Python3 code to demonstrate

005 | # Suffix List Sum

006 | # using list comprehension + sum() + list slicing

007 |

008 | # initializing list

009 | test_list = [3, 4, 1, 7, 9, 1]

010 |

011 | # printing original list

012 | print("The original list : " + str(test_list))

013 |

014 | # using list comprehension + sum() + list slicing

015 | # Suffix List Sum

016 | test_list.reverse()

017 | res = [sum(test_list[: i + 1]) for i in range(len(test_list))]

018 |

019 | # print result

020 | print("The suffix sum list is : " + str(res))

021 |

022 | # or

023 |

024 | def suffix_sum(lst):

025 | result = []

026 | sum = 0

027 | for i in range(len(lst) - 1, -1, -1):

028 | sum += lst[i]

029 | result.append(sum)

030 | return result

031 |

032 |

033 | # input sample list

034 | test_list = [3, 4, 1, 7, 9, 1]

035 | print("The original list :", test_list)

036 |

037 | res = suffix_sum(test_list)

```
038 |  
039 | # printing list  
040 | print("The suffix sum list is :", res)  
041 |  
042 | # or  
... 94 later source lines omitted; inspect the original.
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.